Global convergence and period-three orbits for geodesic operator Sinkhorn

Abstract

We study global convergence of geodesic operator Sinkhorn with a fixed relaxation parameter. For a positivity-improving real Kraus map between matrix sizes $n, m \geq 2$, put $r = \max(m, n)$. Every positive-definite initialization converges projectively when $0 < \omega \leq r/(r-1)$ and $r \geq 3$. For square size two, the sharp universal interval is $0 < \omega < 2$; for square size three, it is $0 < \omega \leq 3/2$. The positive proof uses the trace potential on determinant-one representatives, an exact noncommuting spectral profile, and strict descent at the endpoint. For every fixed $3/2 < \omega < 2$, we construct a positivity-improving 3×3 Kraus map and a prescribed start whose normalized orbit approaches a three-cycle that attracts a neighborhood within the diagonal invariant subspace. One instance has entirely rational map and initial data. The higher-dimensional sufficient bound is not claimed optimal.

Keywords: operator scaling; Sinkhorn iteration; overrelaxation; matrix geometric mean; Lyapunov functions; periodic orbits

2020 Mathematics Subject Classification. Primary 65F35; Secondary 15B48, 39A23, 37B25.

1 The recurrence and the global range

Let $m, n \geq 2$, and let $A_1, \ldots, A_s \in \mathbb{R}^{m \times n}$. Define the completely positive map and its trace adjoint by

$$\Phi(Y) = \sum_{i=1}^{s} A_i Y A_i^T, \qquad \Phi^*(X) = \sum_{i=1}^{s} A_i^T X A_i. \tag{1}$$

Assume that Φ is *positivity improving*: it maps every nonzero positive-semidefinite matrix to a positive-definite matrix. This condition is stronger than mere existence of operator scalings and is retained throughout.

For positive-definite matrices of the same size, define

$$X \#_\omega Z = X^{1/2}(X^{-1/2} Z X^{-1/2})^\omega X^{1/2}. \tag{2}$$

From arbitrary positive-definite X_0, Y_0, use one fixed $\omega > 0$ in the exact recurrence

$$X_{k+1} = X_k \#_\omega \left(m^{-1}\Phi(Y_k)^{-1}\right), \tag{3}$$

$$Y_{k+1} = Y_k \#_\omega \left(n^{-1}\Phi^*(X_{k+1})^{-1}\right). \tag{4}$$

A scaling solution is a positive-definite pair satisfying $X_* = m^{-1}\Phi(Y_*)^{-1}$ and $Y_* = n^{-1}\Phi^*(X_*)^{-1}$. This is the geodesic version, Algorithm 6 and equation (3.6), of Soma and Uschmajew [1]. It is different from their Euclidean, logarithmic-coordinate and Cholesky-factor relaxations.

Let

$$N_d(Z) = \frac{Z}{(\det Z)^{1/d}}, \qquad \mathcal{D}_d = \{Z \succ 0 : \det Z = 1\},$$
$$U_k = N_m(X_k), \qquad V_k = N_n(Y_k). \tag{5}$$

Normalization is used in the analysis; no normalization step is added to (3)–(4). Hilbert distance is

$$d_H(X, Z) = \log \frac{\lambda_{\max}(X^{-1/2} Z X^{-1/2})}{\lambda_{\min}(X^{-1/2} Z X^{-1/2})}. \tag{6}$$

It is unchanged by independent positive scalar rescalings of X and Z.

Theorem 1. *Put $r = \max(m, n)$. Suppose either*

$$r = 2, \quad 0 < \omega < 2; \qquad \text{or} \qquad r \geq 3, \quad 0 < \omega \leq \frac{r}{r-1}. \tag{7}$$

Then there is a unique determinant-normalized scaling pair $(U_*, V_*) \in \mathcal{D}_m \times \mathcal{D}_n$, and

$$U_k \longrightarrow U_*, \qquad V_k \longrightarrow V_* \tag{8}$$

in matrix norm, from every positive-definite initialization. For any operator-scaling solution (X_*, Y_*) with these normalized representatives,

$$d_H(X_k, X_*) + d_H(Y_k, Y_*) \longrightarrow 0. \tag{9}$$

The interval in (7) is sharp as a universal positive-relaxation range when $m = n = 2$ and when $m = n = 3$. The negative three-dimensional assertion is proved in Theorem 2. No sharpness assertion is made for larger or unequal matrix sizes.

The conclusion concerns the normalized matrices and Hilbert distance. It does not assert convergence of the two unnormalized scalar factors or a convergence rate.

1.1 Relation to the known ranges

The final article [1, Theorem 3.3] proves local convergence for $0 < \omega < 2$. Its global Theorem 3.6 gives

$$0 < \omega < \frac{2}{1 + \sqrt{\Lambda_1 \Lambda_2}}, \tag{10}$$

where $\Lambda_1, \Lambda_2 < 1$ are Hilbert-metric contraction constants of the two solution maps. The paragraph following that theorem leaves larger global relaxation values open. The classical positive matrix-scaling result of Lehmann et al. [2, Theorem 3] has the analogous contraction-based range. The earlier global method of Thibault et al. [3] uses a state-dependent safeguard that ensures decrease of its chosen potential. Their Section 4.1 already notes that the naive fixed-parameter composition can fail to converge. The stable-implementation sequel [4] establishes equivalent implementations and does not supply a larger global interval.

The trace/log-determinant objective and its geodesic exponential representation are already present in [1, Section 2.3 and Appendix A.3]. We use its trace part on determinant-one representatives. The new sufficient range (7) depends on the dimensions, and is complementary to (10). For example, in square dimension d consider

$$\Phi_\varepsilon(Y) = (1-\varepsilon)Y + \frac{\varepsilon}{d}\operatorname{tr}(Y)I, \qquad 0 < \varepsilon < 1. \tag{11}$$

It has Kraus matrices $\sqrt{1-\varepsilon}\, I$ and $\sqrt{\varepsilon/d}\, E_{ij}$. Testing the Hilbert-distance ratio at $I + t\,\mathrm{diag}(1,-1,0,\dots,0)$ as $t \to 0$ gives $\Lambda_1, \Lambda_2 \geq 1-\varepsilon$. Thus the endpoint in (10) is at most $2/(2-\varepsilon)$. At $d = 3$, $\varepsilon = 1/10$, this is at most $20/19$, whereas (7) includes $3/2$. For strongly contractive maps, the older range can instead be larger. The dimension-three construction in Section 6 shows that no universal guarantee can cover any fixed $3/2 < \omega < 2$ in that square size, even though local convergence holds there. The counterexample already belongs to the classical diagonal matrix-scaling subclass. It is distinct from the convergent local oscillations discussed in [2] and from the adaptive safeguard in [3]. Ma [5] constructs an initialized six-cycle for a different matrix-scaling acceleration, Bregman Douglas–Rachford splitting; its cumulative-variable update extrapolates the argument of the ordinary Sinkhorn map and is not the fixed geodesic overrelaxation studied here.

2 Normalization and a compact potential

For positive scalars a, b, direct matrix functional calculus gives

$$(aX) \#_\omega (bZ) = a^{1-\omega} b^\omega (X \#_\omega Z), \tag{12}$$

$$\det(X \#_\omega Z) = (\det X)^{1-\omega} (\det Z)^\omega. \tag{13}$$

Hence the normalized actual sequence satisfies

$$U_{k+1} = U_k \#_\omega \mathcal{A}(V_k), \qquad V_{k+1} = V_k \#_\omega \mathcal{B}(U_{k+1}), \tag{14}$$

where

$$\mathcal{A}(V) = (\det \Phi(V))^{1/m} \Phi(V)^{-1}, \qquad \mathcal{B}(U) = (\det \Phi^*(U))^{1/n} \Phi^*(U)^{-1}. \tag{15}$$

Our potential is

$$\mathcal{P}(U,V) = \operatorname{tr}(U\Phi(V)) = \operatorname{tr}(\Phi^*(U)V), \qquad (U,V) \in \mathcal{D}_m \times \mathcal{D}_n. \tag{16}$$

Lemma 1. *There exists $\alpha > 0$ such that, for all positive-semidefinite inputs,*

$$\Phi(V) \succeq \alpha \operatorname{tr}(V) I_m, \qquad \Phi^*(U) \succeq \alpha \operatorname{tr}(U) I_n. \tag{17}$$

Consequently Φ^* is positivity improving, and every sublevel of $\mathcal{P}$ on $\mathcal{D}_m \times \mathcal{D}_n$ is compact.

Proof. The trace-one positive-semidefinite set is compact, and positivity improvement implies

$$\alpha = \min_{V \succeq 0,\ \operatorname{tr} V = 1} \lambda_{\min}(\Phi(V)) > 0. \tag{18}$$

Homogeneity proves the first inequality in (17). For $U \succeq 0$ and any unit vector v, $v^T \Phi^*(U) v = \operatorname{tr}(U \Phi(vv^T)) \geq \alpha \operatorname{tr} U$, proving the second. Thus

$$\mathcal{P}(U, V) \geq \alpha \operatorname{tr}(U) \operatorname{tr}(V) \geq \alpha mn. \tag{19}$$

A sublevel bounds both traces, because $\operatorname{tr} U \geq m$ and $\operatorname{tr} V \geq n$. For determinant-one positive matrices, an upper trace bound bounds every eigenvalue above and away from zero. The sublevel is therefore closed in a compact subset of the positive-definite cone. ∎

The coordinate minimizers in (15) are exact and unique. Indeed, for $C \succ 0$ of size d and $Z \in \mathcal{D}_d$,

$$\operatorname{tr}(CZ) \geq d(\det C)^{1/d}, \tag{20}$$

with equality precisely when $C^{1/2} Z C^{1/2} = (\det C)^{1/d} I$. Thus the unique minimizer is $(\det C)^{1/d} C^{-1}$. Lemma 1 also shows that $\mathcal{P}$ attains a minimum. Every minimizer is a fixed pair of (15).

We recall why that normalized pair is unique; this is consistent with the existing scaling theorem [1, Theorem 2.5]. For two nonproportional positive-definite matrices Z, W, let $aW \preceq Z \preceq bW$ be the optimal Loewner bounds. Both $Z - aW$ and $bW - Z$ are nonzero and positive semidefinite. Positivity improvement makes their images positive definite, so the optimal output bounds improve strictly. Therefore

$$d_H(\Phi(Z), \Phi(W)) < d_H(Z, W). \tag{21}$$

The same holds for Φ^*. Inversion and rescaling preserve Hilbert distance. Two different normalized fixed pairs would therefore yield a strict decrease of their distance under the composition $\mathcal{B} \circ \mathcal{A}$, contradicting fixedness. If their intermediate images coincide, their second components coincide directly. Thus the fixed pair (U_*, V_*) is unique.

Put $\mathcal{P}_* = \mathcal{P}(U_*, V_*)$. At this pair, coordinate optimality and trace adjointness imply

$$\mathcal{P}_* = m(\det \Phi(V_*))^{1/m} = n(\det \Phi^*(U_*))^{1/n}. \tag{22}$$

The pair $X_* = U_*/\sqrt{\mathcal{P}_*}$, $Y_* = V_*/\sqrt{\mathcal{P}_*}$ satisfies the original scaling equations. Writing any raw solution as $X = aU_*$, $Y = bV_*$ in the scaling equations forces $ab = 1/\mathcal{P}_*$. Thus all unnormalized solutions are its reciprocal scalar orbit. This identifies the projective limit without assuming convergence of the raw scalar factors.

3 The coordinate profile and its descent

Lemma 2. *Let $C \succ 0$ have size d, let $U \in \mathcal{D}_d$, and put $a = (\det C)^{1/d}$ and $A = aC^{-1}$. Write the eigenvalues of $U^{1/2} C U^{1/2}/a$ as $e^{t_1}, \ldots, e^{t_d}$. Then $\sum_i t_i = 0$, and for every real*

ω,

$$\operatorname{tr}\big(C(U \#_\omega A)\big) = a\sum_{i=1}^{d} e^{(1-\omega)t_i}, \qquad \operatorname{tr}(CU) = a\sum_{i=1}^{d} e^{t_i}. \tag{23}$$

No commutativity assumption on C and U is needed.

Proof. Set $M = U^{1/2}CU^{1/2}$. Its determinant is $\det C = a^d$. Directly,

$$\begin{aligned} U^{-1/2}AU^{-1/2} &= aM^{-1}, \qquad U \#_\omega A = a^\omega U^{1/2}M^{-\omega}U^{1/2}, \\ \operatorname{tr}\big(C(U \#_\omega A)\big) &= a^\omega \operatorname{tr}(M^{1-\omega}). \end{aligned} \tag{24}$$

The eigenvalue representation gives (23). This calculation uses powers of the single positive matrix M and cyclicity of the trace, so it is valid when C and U do not commute. ∎

Lemma 3. *Let $d \geq 2$, let $t \in \mathbb{R}^d$ satisfy $\sum_i t_i = 0$, and let $-1/(d-1) \leq q < 1$. Then*

$$\sum_i e^{qt_i} \leq \sum_i e^{t_i}. \tag{25}$$

The inequality is strict for $t \neq 0$, except when $d = 2$ and $q = -1$. Moreover, with $\rho_d(q) = \max\{q, -(d-1)q\}$,

$$\sum_i e^{qt_i} - d \leq \rho_d(q)\left(\sum_i e^{t_i} - d\right). \tag{26}$$

Proof. The matrix

$$B_d(q) = qI + \frac{1-q}{d}\mathbf{1}\mathbf{1}^T \tag{27}$$

is nonnegative and doubly stochastic, and $B_d(q)t = qt$. Jensen's inequality in each row, followed by the column-sum identity, proves (25). If $-1/(d-1) < q < 1$, all entries of $B_d(q)$ are positive. Equality then forces all t_i to be equal, hence zero. At $q = -1/(d-1)$, each row averages every coordinate except its own. For $d \geq 3$, equality in every row again forces all coordinates to be equal. For $d = 2$, the endpoint matrix exchanges the two coordinates, so equality holds.

If $0 \leq q < 1$, scalar convexity gives $e^{qt_i} \leq qe^{t_i} + 1 - q$, proving (26). If $q \leq 0$, put $u = -(d-1)q \in [0,1]$. Then

$$\sum_i e^{qt_i} \leq u\sum_i e^{-t_i/(d-1)} + (1-u)d \leq u\sum_i e^{t_i} + (1-u)d, \tag{28}$$

which is the remaining case. ∎

Apply these lemmas with $q = 1 - \omega$ to either coordinate block of (14). If $r \geq 3$ and $\omega \leq r/(r-1)$, then a block of size $d \leq r$ satisfies $q \geq -1/(d-1)$. At the included upper endpoint, a block of size $d = r$ has the strict Jensen endpoint from Lemma 3, while a smaller block lies strictly inside its allowed interval. When $r = 2$, the excluded value $\omega = 2$ is exactly the equality exception in that lemma. Thus, throughout (7), each coordinate update strictly decreases $\mathcal{P}$ unless the coordinate already equals its minimizer.

For the interior range $0 < \omega < r/(r-1)$, define

$$\Delta_X(U,V) = \mathcal{P}(U,V) - m(\det \Phi(V))^{1/m}, \tag{29}$$

$$\Delta_Y(U,V) = \mathcal{P}(U,V) - n(\det \Phi^*(U))^{1/n}, \tag{30}$$

$$c_r(\omega) = \min\{\omega, r-(r-1)\omega\} > 0. \tag{31}$$

Equation (26) yields the useful full-step estimate

$$\mathcal{P}(U_k,V_k) - \mathcal{P}(U_{k+1},V_{k+1}) \geq c_r(\omega)\big(\Delta_X(U_k,V_k) + \Delta_Y(U_{k+1},V_k)\big). \tag{32}$$

The coefficient uses $r \geq d$ for both blocks. At the higher-rank endpoint $\omega = r/(r-1)$, this particular coefficient is zero. We will use the strict coordinate decrease directly there.

4 Convergence through the actual alternating map

For a fixed admissible ω, set

$$D_X(U,V) = \mathcal{P}(U,V) - \mathcal{P}(U \#_\omega \mathcal{A}(V), V), \tag{33}$$

$$D_Y(U,V) = \mathcal{P}(U,V) - \mathcal{P}(U, V \#_\omega \mathcal{B}(U)). \tag{34}$$

These are continuous nonnegative functions. By Section 3,

$$D_X(U,V) = 0 \iff U = \mathcal{A}(V), \qquad D_Y(U,V) = 0 \iff V = \mathcal{B}(U). \tag{35}$$

The actual full-step change is exactly

$$\mathcal{P}(U_k,V_k) - \mathcal{P}(U_{k+1},V_{k+1}) = D_X(U_k,V_k) + D_Y(U_{k+1},V_k). \tag{36}$$

The normalized iterates and their intermediate half-steps stay in the initial compact sublevel of $\mathcal{P}$. Their potential decreases to a finite limit, so both nonnegative terms in (36) tend to zero.

Take any convergent subsequence $(U_{k_j},V_{k_j}) \to (\bar{U},\bar{V})$. Continuity and (35) give $\bar{U} = \mathcal{A}(\bar{V})$. Applying the actual first update, rather than treating successive subsequences independently, yields

$$U_{k_j+1} = U_{k_j} \#_\omega \mathcal{A}(V_{k_j}) \longrightarrow \bar{U} \#_\omega \bar{U} = \bar{U}. \tag{37}$$

The second term in (36) now gives $\bar{V} = \mathcal{B}(\bar{U})$. Every cluster point is therefore the unique normalized fixed pair. Compactness proves convergence of the whole normalized sequence in matrix norm. Equation (9) follows from continuity and scalar invariance of Hilbert distance.

This proof also covers the higher-rank endpoint, using (35). It does not divide by $c_r(\omega)$ there. In the strict interior, telescoping (32) additionally gives

$$\sum_{k=0}^{\infty}\big(\Delta_X(U_k,V_k) + \Delta_Y(U_{k+1},V_k)\big) \leq \frac{\mathcal{P}(U_0,V_0) - \mathcal{P}_*}{c_r(\omega)}. \tag{38}$$

No endpoint summability conclusion is inferred from this estimate.

5 Elementary boundary examples

Consider the real square Kraus family consisting of all matrix units E_{ij}, $1 \le i, j \le d$. It gives the positivity-improving, self-adjoint map

$$\Phi(Y) = \operatorname{tr}(Y)I, \qquad \Phi^*(X) = \operatorname{tr}(X)I. \tag{39}$$

Both normalized coordinate targets are I. If $Y_0 = I$ and $U_0 = N_d(X_0)$, the exact normalized trajectory is

$$U_k = U_0^{(1-\omega)^k}, \qquad V_k = I. \tag{40}$$

This follows by normalizing the actual raw recurrence; it does not prescribe a state at a later time.

For $d = 2$, take $X_0 = \operatorname{diag}(e, e^{-1})$. At $\omega = 2$ the sequence alternates between X_0 and X_0^{-1}. For $\omega > 2$,

$$d_H(U_k, I) = 2|1 - \omega|^k \longrightarrow \infty. \tag{41}$$

This proves the size-two sharpness assertion in Theorem 1.

For $d \ge 3$, the boundary $d/(d-1)$ has a different meaning. Take

$$X_0 = \operatorname{diag}(e^t, \dots, e^t, e^{-(d-1)t}), \qquad t > 0, \quad Y_0 = I. \tag{42}$$

Then

$$\mathcal{P}(U_0, I) = d\big((d-1)e^t + e^{-(d-1)t}\big), \tag{43}$$

$$\mathcal{P}(U_1, I) = d\big((d-1)e^{(1-\omega)t} + e^{(d-1)(\omega-1)t}\big). \tag{44}$$

For every $\omega > d/(d-1)$, the last exponential in (44) eventually dominates (43) as $t \to \infty$, so the first potential change is an increase. Nevertheless, whenever $0 < \omega < 2$, formula (40) still converges to (I, I). Thus this example limits the global monotonicity of the potential but is not an algorithmic counterexample in the omitted interval below two. The next section supplies a different map with an actual nonconvergent orbit in square size three.

6 A three-cycle in square size three

The sufficient interval in dimension three is also necessary for a guarantee over all positivity-improving maps and all positive-definite starts. The obstruction already occurs in the diagonal invariant subclass, which is the classical positive matrix-scaling recurrence [2].

Theorem 2. *For every fixed $3/2 < \omega < 2$, there is a fixed positivity-improving real 3×3 Kraus map with rational Kraus entries and a prescribed positive-definite initial pair*

whose determinant-normalized iterates under (3)–(4) converge phasewise to a nonconstant period-three orbit. At $\omega = 7/4$, the initial matrices can also be chosen rational. Consequently the sharp universal positive-relaxation range in square size three is

$$0 < \omega \le 3/2. \tag{45}$$

Attraction is proved within the diagonal invariant subspace. No claim of uniqueness of all periodic orbits, or of periodicity of the unnormalized scalar factors, is made.

6.1 The fixed family and its logarithmic state

Fix $3/2 < \omega < 2$, and define

$$\begin{gathered} \nu = 2 - \omega, \qquad D = \omega^2 - 3\omega + 3, \qquad \mu = \frac{2\omega - 3}{D}, \\ N = 2\left\lceil \frac{10 + \log_2(1/\nu)}{2\mu} \right\rceil, \qquad \varepsilon = 2^{-N}. \end{gathered} \tag{46}$$

Here $0 < \nu < 1/2$, $3/4 < D < 1$, $0 < \mu < 1$, and N is a positive even integer. Set

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad R = P^2, \quad K = P + \varepsilon(\mathbf{1}\mathbf{1}^T - P) = \begin{pmatrix} \varepsilon & \varepsilon & 1 \\ 1 & \varepsilon & \varepsilon \\ \varepsilon & 1 & \varepsilon \end{pmatrix}. \tag{47}$$

The nine Kraus matrices $A_{ij} = \sqrt{K_{ij}} E_{ij}$ give

$$\Phi(Y) = \operatorname{diag}(K \operatorname{diag} Y), \qquad \Phi^*(X) = \operatorname{diag}(K^T \operatorname{diag} X), \tag{48}$$

where $\operatorname{diag} Y$ denotes the diagonal vector in the right-hand expressions. Their nonzero entries are 1 or $2^{-N/2}$ and hence rational. For every positive-semidefinite input,

$$\Phi(Y) \succeq \varepsilon \operatorname{tr}(Y) I, \qquad \Phi^*(X) \succeq \varepsilon \operatorname{tr}(X) I. \tag{49}$$

Thus positivity improvement holds on the entire cone. The common row and column sum of K is $1 + 2\varepsilon$, so the normalized scaling pair is (I, I). A raw solution is $X_* = Y_* = [3(1 + 2\varepsilon)]^{-1/2} I$. Uniqueness follows from Section 2.

Diagonal starts stay diagonal. Write their centered base-two logarithms as

$$z = \log_2 \operatorname{diag} N_3(X), \quad w = \log_2 \operatorname{diag} N_3(Y), \quad H = \{v \in \mathbb{R}^3 : \mathbf{1}^T v = 0\}, \quad C = I - \tfrac{1}{3}\mathbf{1}\mathbf{1}^T. \tag{50}$$

All vector logarithms and powers are componentwise. Define

$$G(v) = C \log_2(K 2^v), \qquad J(v) = C \log_2(K^T 2^v). \tag{51}$$

Centering the actual diagonal updates, including their scalar factors $1/3$, gives precisely

$$T(z, w) = (z^+, w^+), \qquad z^+ = (1 - \omega) z - \omega G(w), \quad w^+ = (1 - \omega) w - \omega J(z^+). \tag{52}$$

In particular, the second evaluation uses the new first coordinate. Directly,

$$K^T R = K, \qquad KR = RK, \qquad G(Rv) = RG(v), \qquad J(Rv) = G(v). \tag{53}$$

6.2 Three invariant phase boxes

Set

$$p = \frac{(\omega(\omega-3),\ \omega^2,\ -\omega(2\omega-3))}{3D}, \qquad c_i = NR^i p, \qquad b = \frac{N}{3}(-1,-1,2), \tag{54}$$

and let $b_i = R^i b$. Indices i for the phase centers are taken modulo three. With the product max norm on $H \times H$, define

$$B_i = \{(z,w) \in H \times H : \max(\|z - c_i\|_\infty, \|w - c_i\|_\infty) \le 1\}. \tag{55}$$

The following whole-box estimates are the main point of the construction.

Lemma 4. *For every $i = 0,1,2$, the image $T(B_i)$ has radius at most $\omega/2$ around (c_{i+1}, c_{i+1}). On each B_i, the full map T is Lipschitz with constant at most $\omega/2 < 1$. The three boxes are pairwise disjoint.*

Proof. Number the entries of p as p_1, p_2, p_3. Direct algebra gives

$$\begin{gathered} \mathbf{1}^T p = 0, \qquad c_{i+1} = (1-\omega)c_i - \omega b_i, \\ p_2 - p_1 - 1 = \frac{(3-\omega)(\omega-1)}{D} > \mu, \qquad p_2 - p_3 - 1 = \mu, \\ p_1 < p_3 < p_2, \qquad p_2 - p_1 = \frac{\omega}{D} = 2 + \mu\nu > 2. \end{gathered} \tag{56}$$

The strict inequality follows because the difference from μ is $\omega\nu/D$. At $v = c_0$, the second input coordinate is the dominant term in every row of $K2^v$. The three exponent rows are

$$(v_1 - N, v_2 - N, v_3), \quad (v_1, v_2 - N, v_3 - N), \quad (v_1 - N, v_2, v_3 - N). \tag{57}$$

Equation (56) shows that every dominant-minus-competing exponent is at least $N\mu$. In the entire ball $\|v - c_0\|_\infty \le 1$, this margin loses at most two. Hence each of the two competing terms in any row has ratio at most

$$s = 2^{-N\mu+2} \le \frac{\nu}{256} \tag{58}$$

to its dominant term. The last inequality follows from (46).

In each row, extract the dominant term before taking its logarithm:

$$\log_2(K2^v)_j = v_2 + \log_2 K_{j2} + e_j(v), \qquad 0 \le e_j(v) \le \log_2(1+2s) < 4s. \tag{59}$$

The centered logarithm of the second column of K is b. Centering the vector e has max norm at most its coordinate range. Therefore

$$\|G(v) - b\|_\infty \le 4s \qquad (\|v - c_0\|_\infty \le 1). \tag{60}$$

The uncentered logarithm has derivative equal to a row-stochastic softmax matrix $S(v)$. Each row places at most $2s$ probability outside the second coordinate. Thus, with e_2 the second coordinate vector,

$$\|S(v) - \mathbf{1}e_2^T\|_\infty \le 4s, \qquad \|DG(v)\|_\infty \le \|C\|_\infty\, 4s = \tfrac{16}{3}s < 6s. \tag{61}$$

Here $C\mathbf{1}e_2^T = 0$ and $\|C\|_\infty = 4/3$; the matrix norm is the induced max norm. Convexity of the balls gives the corresponding Lipschitz estimates. By (53), on the ball at c_i, G differs from b_i by at most $4s$ and has derivative norm at most $6s$. On the ball at c_{i+1}, the same statements hold for J with target b_i.

Put $a = \omega - 1 = 1 - \nu$. For $(z, w) \in B_i$, the first update and (56) give

$$\|z^+ - c_{i+1}\|_\infty \le a + 4\omega s \le 1 - \nu + \frac{\nu}{32} < 1 - \frac{\nu}{2} = \frac{\omega}{2} < 1. \tag{62}$$

The actual z^+ is consequently in the ball where the estimate for J holds. Applying it in the second update gives the identical bound for $\|w^+ - c_{i+1}\|_\infty$.

For two inputs in B_i at product distance δ, put $h = 6\omega s \le 3\nu/64$. The actual alternating dependence gives

$$\|\Delta z^+\|_\infty \le (a + h)\delta, \qquad \|\Delta w^+\|_\infty \le \big(a + h(a + h)\big)\delta. \tag{63}$$

Since $a + h \le 1 - 61\nu/64 < 1$, both coefficients are at most $a + h \le \omega/2$. This proves contraction on each phase box. Finally, for $i \ne j$ the cyclic permutations satisfy

$$\|c_i - c_j\|_\infty = N(p_2 - p_1) = \frac{N\omega}{D} > 2N > 2, \tag{64}$$

so the boxes are disjoint. ∎

6.3 The initialized nonconvergent sequence

The complete metric space B_0 is invariant under T^3, whose contraction constant is at most $(\omega/2)^3$. It therefore has a unique fixed point q_0 in that box. The points $q_1 = Tq_0$ and $q_2 = T^2 q_0$ lie in B_1 and B_2 and satisfy $Tq_2 = q_0$. They are distinct by (64). Every orbit starting in B_0 converges to them along its three return subsequences, with return contraction factor at most $(\omega/2)^3$. This establishes uniqueness of the cycle *within these boxes.*

In particular, choose the fixed initial pair

$$X_0 = Y_0 = \operatorname{diag}(2^{c_{0,1}}, 2^{c_{0,2}}, 2^{c_{0,3}}). \tag{65}$$

It is determinant one and has log state $(c_0, c_0) \in B_0$. Thus the conclusion concerns the actual initialized recurrence, rather than a periodic point chosen implicitly as the starting

state. The coordinate phase balls themselves are disjoint, so both normalized matrix sequences have three distinct subsequential limits. Moreover, for every $k \geq 0$,

$$d_H(X_k, I) = (\log 2)\big(\max_j z_{k,j} - \min_j z_{k,j}\big) \geq \left(\frac{N\omega}{D} - 2\right) \log 2 > 0. \tag{66}$$

No conclusion about the raw scalar gauges enters this argument.

For an entirely rational example, take $\omega = 7/4$. Then

$$\nu = \frac{1}{4}, \quad D = \frac{13}{16}, \quad \mu = \frac{8}{13}, \quad N = 20, \qquad c_0 = \frac{1}{39}(-700, 980, -280). \tag{67}$$

Use $\varepsilon = 2^{-20}$ in (47) and replace (65) by

$$X_0 = Y_0 = \operatorname{diag}(2^{-18}, 2^{25}, 2^{-7}). \tag{68}$$

Its centered log vector differs from c_0 by $(-2, -5, 7)/39$, so it belongs to the required ball. The same full-orbit conclusion applies.

Theorem 1 proves the positive side of (45). The construction above excludes every $3/2 < \omega < 2$. For $\omega \geq 2$, use (39) in size three with $X_0 = \operatorname{diag}(e, e^{-1}, 1)$ and $Y_0 = I$; formula (40) gives nonconvergence. This completes Theorem 2.

The cycle attracts the stated diagonal phase boxes and is far from the scaling solution and does not conflict with the known local convergence for every $0 < \omega < 2$ [1, 2]. The data are fixed along the orbit, the Kraus map is positivity improving on all positive-semidefinite inputs, and no restart or later free initialization is used. The sharp conclusion here concerns square size three. Optimality for larger sizes or unequal sizes with maximum three is left open.

References

[1] T. Soma and A. Uschmajew, *Accelerating operator Sinkhorn iteration with overrelaxation*, Mathematical Programming (2026), published online 26 May 2026. doi:10.1007/s10107-026-02361-1.

[2] T. Lehmann, M.-K. von Renesse, A. Sambale and A. Uschmajew, *A note on overrelaxation in the Sinkhorn algorithm*, Optimization Letters **16** (2022), 2209–2220. doi:10.1007/s11590-021-01830-0.

[3] A. Thibault, L. Chizat, C. Dossal and N. Papadakis, *Overrelaxed Sinkhorn–Knopp algorithm for regularized optimal transport*, Algorithms **14** (2021), 143. doi:10.3390/a14050143.

[4] H. Eisenmann, T. Soma, X. Tang and A. Uschmajew, *Numerically stable variants of overrelaxation for operator Sinkhorn iteration*, arXiv:2603.12955v1, 13 March 2026. arXiv:2603.12955v1.

[5] S. Ma, *BDRS can fail for matrix scaling and optimal transport*, Optimization Online preprint, posted 30 September 2026 (manuscript dated 28 September 2026). Optimization Online.